Alec D. Wendland

Curriculum Vitae

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Research Interests

Scientific computing, applied mathematics, numerical PDEs/ODEs, numerical analysis, computational physics, computational materials science

Education

The University of Connecticut — **PhD Applied Mathematics** August 2021 – May 2027 (expected)
Storrs, CT *GPA:* 3.984/4.0

- **Advisor:** Dr. Xiaodong Yan
- **Relevant coursework:** Applied mathematics, numerical analysis, FEM, PDE, ODE, materials science and engineering, transport phenomena, computational physics

Carroll University — **B.S. Mathematics** (Minor: Computer Science) September 2017 – May 2021
Waukesha, WI *GPA:* 4.0/4.0

Publications and Manuscripts

1. A. D. Wendland and X. Yan, *Polar director structure of $SmAP_F$ phase of bent-core liquid crystals in thin planar cells with bias electric field*, Journal of Mathematical Analysis and Applications, to appear. arXiv:2606.22668 [math.AP].
2. T. E. St. George and A. D. Wendland, *Orthonormalization of a subset of Bernstein polynomial basis functions*, PUMP Journal of Undergraduate Research **7** (November 2024), 258-273, doi:10.46787/pump.v7i0.3765.

Manuscripts in Preparation

1. A. D. Wendland and X. Yan, *Well-posedness and long-time behavior of a one-dimensional Landau–Khalatnikov equation for $SmAP_F$ bent-core liquid crystals*.

Research Experience

The University of Connecticut, Department of Mathematics Storrs, CT
Graduate Researcher *August 2021 – present*

- Developed and analyzed **nonlinear PDE/ODE models for ferroelectric liquid crystals**, deriving equilibrium and evolution equations with physically motivated nonlinear boundary conditions.
- Implemented **finite-difference and variational discretizations** using gradient-descent, Newton, and shooting methods in MATLAB; evaluated convergence, numerical stability, and parameter sensitivity.
- Established **existence and uniqueness results** and derived **parameter thresholds governing equilibrium director alignment**; validated theoretical predictions through numerical simulation.
- Coauthored a peer-reviewed research article accepted for publication in the Journal of Mathematical Analysis and Applications and a second manuscript in preparation; presented research findings at regional and national conferences and technical seminars.

Technical Projects

FCC Metals Molecular Dynamics Simulator GitHub & Web App
Independent Project | Python, C++, pybind11 *November 2025 – present*

- Developed **modular scientific simulation software** in Python and C++ for atomistic modeling of eight FCC metals using Lennard–Jones interactions, periodic boundary conditions, and NVE/NVT ensembles.
- Implemented velocity-Verlet integration, short-range force evaluation, and Verlet neighbor lists; accelerated performance-critical kernels in C++/pybind11, achieving **483–594× faster force evaluation** and **421–463× faster NVE integration** than non-vectorized Python across 108–500 atoms.
- Verified Python/C++ agreement for forces, energy, virial, and trajectories; maintained a **104-test pytest suite** with continuous integration across Python 3.11–3.13.
- Built a **verification and validation suite** assessing timestep convergence, long-time energy and momentum conservation, temperature stability, equipartition, coordination number, and FCC structural behavior.
- Computed RDF, MSD, VACF, diffusion coefficients, and structure factors to **characterize structural and dynamical behavior** across temperature regimes; **integrated analysis with reproducible reports**, 3D visualization using OVITO/PyOVITO, and a deployed Streamlit application.

Technical Skills

Programming: Python, C++, MATLAB, Julia, Java
Scientific Computing: NumPy, SciPy, Matplotlib, pybind11, pytest, Git/GitHub Actions, Jupyter
Numerical Methods: finite differences, finite elements, numerical linear algebra, Newton and gradient methods, shooting methods, time integration, convergence and stability analysis
Computational Modeling: PDE/ODE models, variational models, molecular dynamics, verification and validation, RDF/MSD/VACF and diffusion analysis, OVITO/PyOVITO

Leadership and Teaching Experience

The University of Connecticut, Department of Mathematics — Storrs, CT

Calculus II TA Lead — *Fall 2026*

- Coordinated coursewide communication and instructional support for Calculus II and cross-course TA scheduling for the calculus sequence.

Lead Instructor and Teaching Assistant — *August 2021 – present*

- Instructor of record for nine undergraduate mathematics courses, including differential equations, technical writing, and business calculus; teaching assistant for the calculus sequence.

Vice President and Treasurer | SIAM Graduate Student Chapter — *August 2021 – present*

- Led chapter operations and collaborated with faculty and graduate students to organize technical workshops, seminars, and professional events.

Graduate Mentor | Directed Reading Program — *Fall 2024*

- Independently advised an undergraduate student through a semester-long mathematical research project.

Presentations

1. *Well-posedness and long-time behavior of a one-dimensional Landau-Khalatnikov model for ferroelectric bent-core liquid crystals*, SIAM Conference on Mathematical Aspects of Materials Science, Bellevue, WA, May 2027 (*forthcoming*).
2. *Gradient Flows of Energy Functionals*, SIGMA Seminar, UConn, May 2026 (Talk).
3. *Polar Director Structures in Bent-Core Liquid Crystals Under an Electric Bias Field*, FADE Conference, Fairfield University, April 2026 (Talk).
4. *Polar Director Structures in Bent-Core Liquid Crystals Under an Electric Bias Field*, Workshop in PDE and Applied Math in the Northeast Region, UConn, March 2026 (Talk).
5. *The Direct Method in the Calculus of Variations*, Mathematics Continued Conference, UConn, March 2024 (Talk).
6. *Variational Problems in Liquid Crystal Theory*, Geometric Analysis Student Seminar, UConn, October 2023 (Talk).
7. *Negative Sobolev Spaces and a Compactness Result in Liquid Crystal Theory*, Geometric Analysis Student Seminar, UConn, March 2023 (Talk).
8. *Two Proofs of an Explicit Orthonormalization Formula for Bernstein Polynomials*, Celebrate Carroll Academic Research Conference, Carroll University, April 2020 (Talk).
9. *Bernstein Polynomial Reproducing Kernel Method*, Celebrate Carroll Academic Research Conference, Carroll University, April 2019 (Talk).
10. *Bernstein Polynomial Reproducing Kernel Method*, Joint Mathematics Meetings, Undergraduate Poster Session, Baltimore, MD, January 2019 (Poster).

Awards and Honors

1. Predoctoral Fellowship, UConn, Spring 2025 — $8,078
2. Summer Research Assistantship, UConn, Summer 2024 — $5,498
3. Summer Fellowship, UConn, Summer 2022 — $5,000
4. Alan and Linda Thompson Scholarship, Carroll University, August 2020 — $2,500
5. Mathematics Major of the Year Award, Carroll University, 2019, 2021
6. Pioneer Scholar Grant, Carroll University, Summer 2018 — $3,000
7. National Merit Scholarship, Mead Witter Foundation, awarded annually 2017–2020 — $20,000 total
8. Wisconsin Academic Excellence Scholarship, awarded annually 2017-2020 — $10,000 total